

Evaluating Bias in Segmentation Metrics: The Role of Anatomical Structure Size in MRI Brain Imaging

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Abstract— Accurate evaluation of segmentation models in medical imaging is critical for their clinical applicability. However, the size of anatomical structures might influence the result of certain evaluation metrics. In this study, segmentation models tailored for small and large brain structures as well as a model indifferent to structure sizes are investigated to reveal metric-specific biases with respect to segmentation size. The performance of different evaluation metrics is compared over the amygdala, the thalamus, the hippocampus, white and grey matter. Statistical tests reveal that Dice coefficient, volumetric similarity and Rand index are more sensitive to structure size, whereas AUC and Hausdorff distance are less influenced. The findings highlight the importance of size-appropriate metric selection.

Keywords—medical image segmentation, MRI brain imaging, segmentation evaluation metrics, anatomical structure size bias, metric-specific analysis

I. INTRODUCTION

The use of artificial intelligence (AI) in medical image analysis is rapidly growing, with segmentation, traditionally performed by clinicians, now increasingly handled by AI algorithms. These systems reduce the time needed to inspect medical scans, minimize inter-operator variability, and achieve diagnostic accuracy comparable to clinicians, as supported by a growing body of research.

Despite these advancements, reliable evaluation of AI models remains challenging. A troubling trend of cherry-picking or misusing metrics to inflate performance scores has emerged [1], introducing bias and undermining credibility. Robust evaluation is critical to ensure these systems' reliability in clinical settings.

Semantic segmentation evaluation involves assessing both classification accuracy and localization correctness. Many metrics exist for this purpose, and Figure 1 highlights some and their relative frequency about brain segmentation research, reflecting their popularity and relevance.

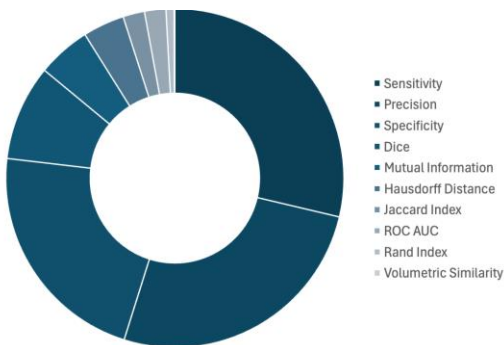


Fig. 1. Relative frequency of metrics used in articles published in 2024 for evaluation of brain segmentation, as sourced from Google Scholar

The metrics can be grouped into six categories, as proposed by Taha et al. [2]: overlap-based, volume-based, pair-counting, information-theoretic, probabilistic, and spatial distance-based. Overlap-based metrics are the most common. These compare the overlap between segmented regions and ground truth and include metrics like Dice, Jaccard, sensitivity, and specificity. All these can be derived from the confusion matrix. Volume-based metrics focus on the volume of segmented regions, ignoring shape differences. Volumetric similarity, a key metric here, measures volume differences and is less sensitive to boundary errors, but it can yield high values even without overlap. Pair-counting metrics, such as the Rand index and adjusted Rand index, assess agreement by counting point pairs in segmentations. Information-theoretic metrics, including mutual information and variation of information, quantify shared information, useful for comparing segmentations with varying labels or intensity distributions. Probabilistic measures, like interclass correlation, Cohen's Kappa, and area under the curve (AUC), account for chance agreement, offering robustness in variable segmentations. Finally, spatial distance-based metrics, like Hausdorff distance and average distance, measure boundary accuracy but are sensitive to outliers.

II. METHODS

A. Data

The dataset consists of 3 Tesla MRI scans from 30 healthy subjects in the Human Connectome Project (HCP) [3], including T1-weighted (T1w) and T2-weighted (T2w) volumes. Both modalities are in the native T1w subject-space and have undergone bias field correction. The images are not skull-stripped but are defaced for anonymity. Ground truth annotations, including label maps and brain masks, were generated using FreeSurfer 5.3 [4], a tool that provides "silver-standard" labels with some imperfections. These annotations are commonly used in medical image analysis due to the limited availability of expert manual annotations. The dataset also includes an affine transformation for aligning the images to a standardized atlas space. The training set comprises 20 subjects, while the testing set includes 10 subjects.

B. Segmentation Pipeline

a) Pre-processing

Preprocessing aims to enhance the quality of the input images and prepare them for image segmentation. First, image normalization was performed to standardize image intensities across the dataset. The intensity values were normalized by subtracting the mean and dividing by the

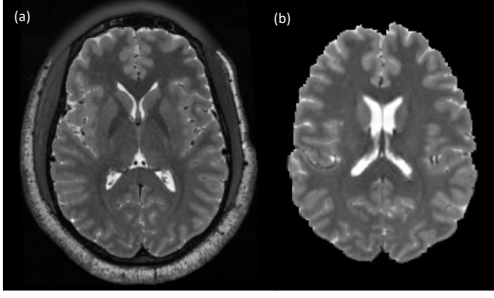


Fig. 2. MRI-image before (a) and after (b) pre-processing

standard deviation. This ensured consistency in intensity ranges, reducing variations that could negatively affect downstream processing. Next, skull stripping was applied to remove non-brain tissues using a binary brain mask. Finally, image registration aligned the input images to a reference atlas using a pre-computed transformation. Linear interpolation was applied for the registration of continuous intensity data, while nearest-neighbour interpolation was used for discrete data such as ground truth segmentation masks. This step ensured that all images were spatially aligned for consistent analysis. Fig. 2 shows the effect of preprocessing.

b) Random Forest Classifier

The brain tissue segmentation is performed using a random forest classifier, a supervised machine learning method based on an ensemble of decision trees. The random forest assigns a class label to each pixel. The possible classes consist of background, white matter (cortical and cerebellar), grey matter, hippocampus, amygdala and thalamus. Features such as intensity-based information, spatial coordinates, and gradient intensity values are extracted. A feature matrix and corresponding label vector are created as input of the random forest classifier.

As this research focuses on the performance of evaluation metrics on both small and large brain structures, three separate random forest classifiers are trained, one tailored on large, one on small structures and one indifferent to structure sizes. This approach aims to uncover potential metric-specific biases and highlight the importance of selecting metrics that align with the size and characteristics of the segmented regions.

Rather than excluding labels, all anatomical structures were retained within the random forest model. The model's Dice coefficients were optimized for specific subsets—small structures such as the hippocampus, amygdala, and thalamus, and larger structures like white and grey matter. This targeted optimization allowed for an evaluation of how other metrics respond when the model is fine-tuned for specific structure sizes.

All models were implemented using the scikit-learn library for Python [8], employing its *RandomForestClassifier* for constructing the random forest models and *GridSearchCV* for hyperparameter optimization.

The grid search employed a 5-fold cross-validation approach to ensure robust evaluation, using the multi-class dice similarity coefficient as the scoring metric. The parameter space for the search included variations in the number of decision trees and the maximum depth of each tree. The Dice coefficient was calculated as an average across all classes, without weighing individual classes differently, to prevent prioritizing one class over another.

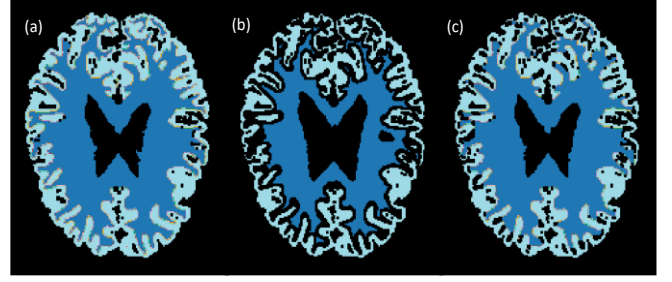


Fig. 3. Segmentation mask (a) after selecting only largest components, (b) after erosion and (c) after dilation

The grid search for all three models was performed using a parameter grid that included 40, 50, 60, 70, 80, and 100 for both the number of estimators and the maximum depth. The optimized parameters were as follows: for the small structure-optimized model, a maximum depth of 100 and 80 estimators; for the large structure-optimized model, a maximum depth of 40 and 100 estimators; and for the structure-size-independent model, a maximum depth of 80 and 60 estimators.

c) Post-processing

To enhance the quality of the segmentation results, post-processing was implemented to remove noise, refine boundaries, and ensure anatomical consistency. First, the most relevant regions for each segmented class were isolated. For most structures, only the largest connected region was retained to eliminate small, irrelevant fragments. For white matter, which often includes multiple disjoint regions, the two largest regions were preserved. Each segmentation class was processed individually by creating a mask to extract only that specific class. The selected regions on this mask were refined and integrated back into the final segmentation, as displayed in Figure 3(a).

Following this, boundary smoothing was applied to correct any irregularities. This involved two steps: first, an erosion step to remove small protrusions and create smoother edges (Figure 3(b)). Secondly, a dilation step is performed to restore the overall size of the structures while preserving the corrected boundaries. This is shown in Figure 3(c). These morphological operations ensured that the final segmentation masks were both visually coherent and anatomically feasible.

C. Evaluation of Segmentation

The segmentation results of all models on the test data are then evaluated using different metrics. Ideally, all possible metrics would be investigated but due to the time and resource limitations of this project, one metric per category was selected.

Overlap-based metrics are expected to be most sensitive to segmentation size and are very popular for evaluating segmentations. In this study, the Dice coefficient is investigated. Volumetric similarity, representing the volume-based metrics which are also expected to be influenced by structure size, is also considered. For the pair-based metrics, the Rand index is used. Information-theoretic metrics, which are expected to be less affected by segmentation size, are represented by mutual information. Similarly, for the probabilistic metrics, only AUC is investigated. Finally, for spatial distance-based metrics, Hausdorff distance (modified to select only the 95th percentile for robustness) is

considered, as a less straightforward relationship is expected for this category.

D. Data analysis

Data analysis was performed in Python using the Pandas [8] library to perform data manipulation and basic statistical operations. To create visualisations of the resulting metrics the Matplotlib [6] and Seaborn [7] libraries were used.

The analysis focuses on three key aspects. First, the distribution of metrics per label and per model is visualized for each metric separately, allowing for a detailed comparison of model performance across different labels. Second, a single plot is created to show the relative differences between the small or large structure-optimized models and the indifferent model in all metrics. Finally, statistical tests are performed to examine p-values, assessing whether the metrics in the small structure-optimized model yield significantly different results compared to the indifferent model, and similarly for the large structure-optimized model.

III. RESULTS AND DISCUSSION

A. Distribution of metrics by label and model

To assess the impact of the models on the selected metrics, the distribution of scores for each metric separately across all labels and models is visualized. To avoid overloading the paper with figures, only one example visualization is included (Figure 4), the others can be found in the Appendix. This provides a preliminary insight into which metrics are most influenced by the model for each label. While it is challenging to determine definitively which metrics are most affected based solely on these plots, some trends are already apparent. For instance, mutual information shows minimal variation across models, as expected. Similarly, the Rand index appears to be less impacted by the model, where the differences seem to be less pronounced compared to mutual information. Hausdorff distance appears to be the most impacted metric. Volumetric similarity and the Dice coefficient are also notably affected, particularly for the amygdala and hippocampus. In contrast, AUC seems less impacted by the choice of model.

B. Relative Differences in Metric Performance Across Models

The analysis of mean relative differences in metrics over all labels between the models optimized for small or large structures and the indifferent model reveals several key patterns (Figure 5). Hausdorff distance shows the most

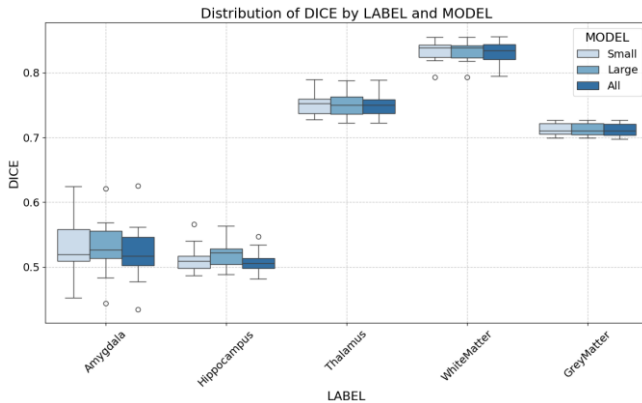


Fig. 4. Boxplots showing the distribution of the Dice coefficients across all labels and models

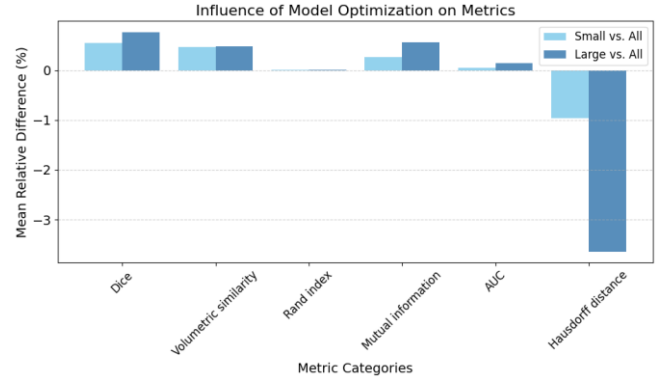


Fig. 5. Mean relative difference between models optimized for small or large structures and an indifferent model, across different metrics

pronounced differences, particularly for the model optimized for large structures. Dice also exhibits notable differences, followed by volumetric similarity. Rand index exhibits the smallest difference, consistent with the observations from the previously discussed distribution. AUC also demonstrates relatively small differences. Interestingly, contrary to expectations, mutual information shows relatively larger differences. This discrepancy can likely be attributed to the use of relative differences, where smaller metric values result in disproportionately larger relative differences. This effect is particularly relevant for mutual information, which has very low baseline values.

C. Statistical Significance of Per-Metric Differences Between Models

Figure 6 illustrates the p-values derived from statistical tests comparing the performance of models optimized for small or large anatomical structures with the size-indifferent model, across various evaluation metrics and labels. Dark blue to light blue cells indicate statistically significant differences ($p < 0.05$) with lighter cells corresponding to higher p-values, white cells represent p-values above 0.05 indicating non-significant results.

Metrics such as the Dice coefficient and volumetric similarity demonstrate significant p-values for most labels for both models. Particularly the smaller structures like the hippocampus and amygdala are affected. The size of these segmented structures changed due to the optimization of the model, leading to significant differences in the evaluation metrics. This outcome is consistent with the fact that overlap- and volume-based metrics are particularly sensitive to small structures, where even minor changes in segmentation can result in substantial variations in the metric values.

Unexpectedly, the Rand index displays significant p-values across many labels for both the small- and large-structure optimized models. In contrast, AUC demonstrates fewer significant results across all labels and models. In the small-label-optimized model, the AUC did not differ significantly for most small structures, except for the thalamus, while a significant difference was also observed for the larger grey matter. Conversely, in the large-label-optimized model, significant differences in AUC were observed for the amygdala, thalamus, and white matter. This shows some correspondence with expectations, as probability-based metrics are generally less influenced by structure size. When significant differences do occur, they can be observed across structures of various sizes.

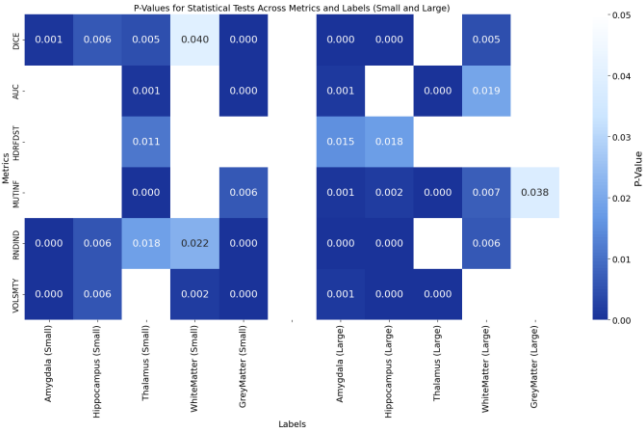


Fig. 6. P-values showing statistical significance between the model optimized on the label and the indifferent model for each metric.

Darker blue cells indicate significant results ($p < 0.05$), lighter blue cells approach the significance threshold, and white cells represent non-significant results ($p > 0.05$).

Interestingly, mutual information, which was anticipated to be the least influenced, shows more significant results than expected, particularly for the large-structure optimized model. Another noteworthy finding is related to the Hausdorff distance. While earlier results highlighted substantial differences for this metric, it appears to be one of the least affected metrics in this figure, showcasing non-significant differences. While the metric might vary across models, those differences are not consistent enough to be deemed statistically significant for certain structures.

Overall, the model optimized for small structures shows fewer significant differences for the amygdala and hippocampus, while the model optimized for large structures exhibits more pronounced significant differences. The same behaviour is observed for grey matter, but the other way around. This contradicts expectations, as one would anticipate that optimizing a model for small or large structures would lead to improved performance for those respective structures. However, by evaluating the effect size, it can be confirmed that these significant changes correspond to meaningful improvements, e.g. as evidenced by the increase in Dice coefficient. This unexpected result may suggest that the model's focus remained too generalized despite the targeted optimization.

IV. LIMITATIONS

The models are evaluated using a limited set of metrics, with only one metric selected from each category. While this approach provides a focused analysis, it may overlook trends that could emerge from a more exhaustive evaluation of all available metrics within each category.

Additionally, the segmentation models employed are based on random forest classifiers. These are robust and interpretable but are often outperformed by newer deep learning approaches. The findings in this research might not translate to more complex models.

Another important limitation of this study is the reliance on generated ground truth segmentations. While these segmentations are widely regarded as "silver standards" in the field, they are not equivalent to manual annotations created by experts. The inherent inaccuracies in these generated segmentations introduce an additional layer of uncertainty into the evaluation process. This, in turn, can influence the

performance metrics, as the evaluation is contingent on comparing model outputs to a potentially flawed reference. Metrics that are highly sensitive to boundary accuracy, such as the Hausdorff distance, are particularly susceptible to such imperfections. Similarly, overlap-based metrics like the Dice coefficient may penalize models unfairly if the ground truth fails to accurately represent the true anatomical boundaries.

Moreover, based on the observed segmented volumes, ensuring biologically plausible ground truth segmentation is essential. Disconnected grey and white matter regions and the absence of basal ganglia segmentation can confuse the model. Our post-processing steps addressed these issues by improving connectivity, though minor leakage into adjacent structures occurred. This trade-off highlights the need for more consistent and accurate ground truth segmentation.

These limitations emphasize the need for caution when interpreting results and suggest that incorporating expert-verified annotations in future studies could enhance the reliability and accuracy of the findings. Additionally, as this study focused solely on analytical significance, future studies could involve a more comprehensive examination of clinical significance guided by expert input.

V. CONCLUSION

This study investigated the biases in evaluation metrics for segmentation models with respect to anatomical structure size. By examining models optimized for small and large brain structures alongside a size-indifferent model, the analysis revealed significant metric-specific behaviors.

The Dice coefficient, being an overlap-based metric, is highly sensitive to small structural differences, especially in smaller structures like the hippocampus and amygdala. This is because small errors in the segmentation of these structures (e.g., a small misclassification) can result in significant changes in the overlap between the predicted and ground truth regions. For smaller structures, these small errors often lead to noticeable fluctuations in the Dice coefficient, making it more sensitive to performance changes in models optimized for smaller anatomical structures. Hausdorff distance may not have shown significant differences across the different models because it is heavily influenced by small mismatches between segmentations, especially at the boundaries of structures. Even a small error in a single segmentation slice could drastically increase the Hausdorff distance, potentially overshadowing the overall improvement in model performance.

Interestingly, metrics like the Rand index also showed significant differences across models, contrary to initial expectations, while probability-based metrics such as AUC exhibited less influence from structure size, aligning with predictions. Mutual information, however, demonstrated more significant differences over the different structures than anticipated, especially for large-structure-optimized models.

The study highlights the importance of selecting appropriate evaluation metrics that align with the characteristics of the segmented structures. Metrics like Dice are significantly more sensitive for smaller structures, whereas AUC showed no consistent significant differences, emphasizing the need for careful metric selection based on structure size. Metrics sensitive to size biases may not always provide reliable performance evaluations, particularly when applied to datasets with a wide range of structure sizes.

REFERENCES

- [1] D. Müller, I. Soto-Rey, and F. Kramer, "Towards a guideline for evaluation metrics in medical image segmentation," *BMC Research Notes*, vol. 15, no. 1, Jun. 2022, doi: 10.1186/s13104-022-06096-y.
- [2] A. A. Taha and A. Hanbury, "Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool," *BMC Medical Imaging*, vol. 15, no. 1, 2015. doi: 10.1186/s12880-015-0068-x.
- [3] D. C. Van Essen *et al.*, "The WU-Minn Human Connectome Project: an overview," *Neuroimage*, vol. 80, pp. 62-79, Oct 15 2013, doi: 10.1016/j.neuroimage.2013.05.041.
- [4] B. Fischl, "FreeSurfer," *Neuroimage*, vol. 62, no. 2, pp. 774-81, Aug 15 2012, doi: 10.1016/j.neuroimage.2012.01.021.
- [5] W. McKinney, "Data Structures for Statistical Computing in Python," *Proceedings of the 9th Python in Science Conference*, 2010, pp. 51–56. Available: <https://doi.org/10.25080/Majora-92bf1922-00a>.
- [6] J. D. Hunter, "Matplotlib: A 2D Graphics Environment", *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90-95, 2007.
- [7] M. L. Waskom, "Seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021. doi: 10.21105/joss.03021.
- [8] F. Pedregosa, *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

APPENDIX

This appendix provides supplementary materials that support the findings discussed in the main text. These boxplots, grouped by metrics such as volumetric similarity, Rand index, mutual information, AUC, and Hausdorff distance, enable a deeper understanding of metric-specific behaviours highlighted in the results section.

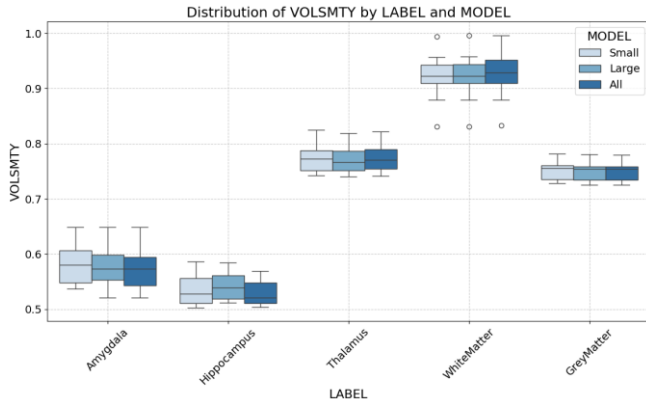


Fig. A.1 Boxplots showing the distribution of the volumetric similarity across all labels and models

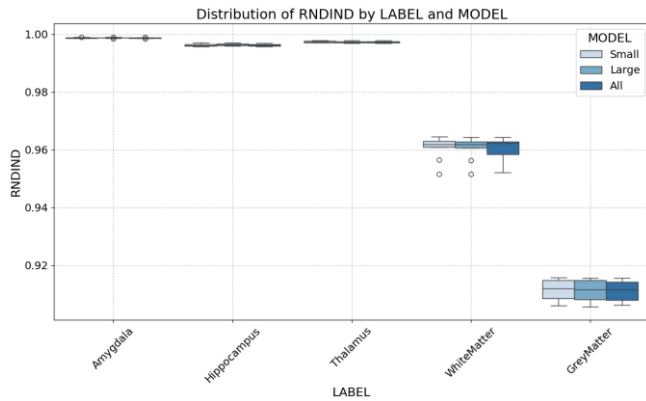


Fig. A.2 Boxplots showing the distribution of the Rand index across all labels and models

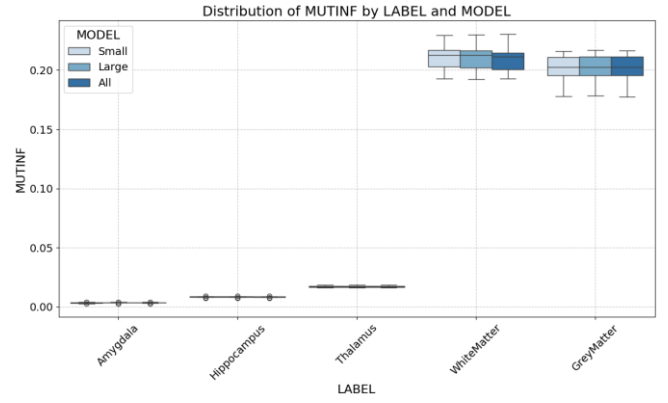


Fig. A.3 Boxplots showing the distribution of the mutual information across all labels and models

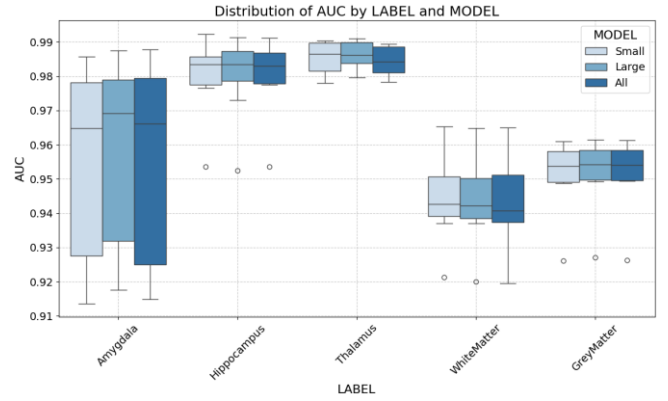


Fig. A.4 Boxplots showing the distribution of the AUC across all labels and models

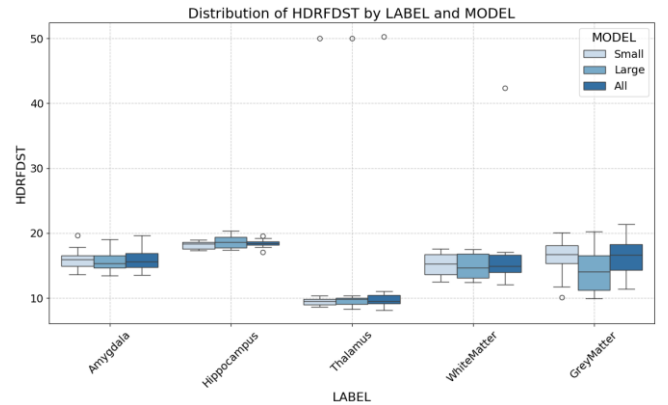


Fig. A.5 Boxplots showing the distribution of the Hausdorff distance across all labels and models